Problem 1 - Fourier Series

Compute the coefficients a_n, b_n in the Fourier series representations of each of the following functions. Then, determine the x at which the series representations fail, if any.

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(\frac{n\pi}{L}x)\,dx$$
$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(\frac{n\pi}{L}x)\,dx$$

(a) $f(x) = 1 + \cos(x) - \sin(2x) \quad L = \pi$

$$\tilde{f}(x) = \frac{b_0}{2} + \sum_{n=1}^{\infty} a_n \sin(nx) + b_n \cos(nx)$$
$$1 + \cos(x) - \sin(2x) = \frac{b_0}{2} + \sum_{n=1}^{\infty} a_n \sin(nx) + b_n \cos(nx)$$

By inspection:

$$\frac{b_0}{2} = 1 \implies b_0 = 2$$
$$b_1 = 1, \quad b_n = 0 \text{ for } n \neq 1$$
$$a_2 = -1, \quad a_n = 0 \text{ for } n \neq 2$$

Since $f(x) = 1+\cos(x)-\sin(2x)$ is smooth and continuous, it's fourier series representation does not fail for any x.

(b) $f(x) = x, \quad L = \pi$

Solving for a_n and b_n using formulas from class.

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin(nx)\,dx$$

Integration by parts (table method):

x	$\sin(nx)$
-1	$-\frac{1}{n}\cos(nx)$
0	$-\frac{1}{n^2}\sin(nx)$

$$
\begin{aligned}
a_n\pi &= -\frac{x}{n}\cos(nx) + \frac{1}{n^2}\sin(nx)\Big|_{-\pi}^{\pi} \\
a_n\pi &= -\frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\sin(n\pi) - \left(-\frac{-\pi}{n}\cos(-n\pi) + \frac{1}{n^2}\sin(-n\pi)\right) \\
a_n\pi &= -\frac{\pi}{n}\cos(n\pi) - \left(-\frac{-\pi}{n}\cos(-n\pi)\right) \\
a_n &= -\frac{1}{n}\cos(n\pi) - \left(-\frac{-1}{n}\cos(-n\pi)\right) \\
a_n &= -\frac{1}{n}\cos(n\pi) - \left(\frac{1}{n}\cos(-n\pi)\right) \quad \cos(n\pi) = \cos(-n\pi) = (-1)^n \\
a_n &= -\frac{1}{n}(-1)^n - \left(\frac{1}{n}(-1)^n\right) \\
a_n &= -\frac{2}{n}(-1)^n
\end{aligned}
$$

Since $f(x) = x$ is an odd function, and $\cos(nx)$ is an even function, $f(x)\cos(nx)$ is an odd function. The integral $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\cos(\frac{n\pi}{\pi}x)\,dx$ would be 0 since it is and odd function integrated over symmetric bounds. Therefor $b_n = 0 \ \forall \ n \in \mathbb{Z}^+$